Section 3.4 The Chain Rule

The Chain Rule is used to find derivatives for the composition of functions.

- $\sqrt{x^2+7}$ is composed of x^2+7 and $\sqrt{x}$
- $\sin(3x)$ is composed of $3x$ and $\sin(x)$

outer function (sin) inner function (3x)

Ex ① Find the derivative of $y = (7x+5)^2$.

we can start by multiplying out $(7x+5)^2$

$$y = 49x^2 + 70x + 25$$

then use the Power Rule

$$y' = 98x + 70$$

factor out 2

$$y' = 2(49x + 35)$$

factor out 7

$$y' = 2(7x+5)\cdot 7$$

Now, compare with the original function

$$y = (7x+5)^2$$

take derivative using Power Rule

And Chain Rule

$$y' = 2\cdot(7x+5)^{2-1}\cdot 7$$

The Chain Rule

$$[f(g(x))]' = \underbrace{f'(g(x))}_{\text{derivative of outer function}} \cdot \underbrace{g'(x)}_{\text{derivative of inner function}}$$

f: outer function

$g(x)$: inner function

Ex ② $y = \sin(x^2+5x+1)$

chain rule

$$y' = \cos(x^2+5x+1)\cdot(2x+5)$$ ✓

Ex ③ $y = e^{\sin x}$ chain rule

$y' = e^{\sin x} \cdot \cos x$

Ex ④ $f(x) = \cos^2 x$ or $(\cos x)^2$

chain rule

$f'(x) = 2 \cdot \cos x \cdot (-\sin x)$

$f'(x) = -2 \cdot \cos x \cdot \sin x$ ✓

Ex ⑤ $h(x) = \sqrt{5x^3 + x^2}$

remember that $\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}}$

$h'(x) = \frac{1}{2\sqrt{5x^3+x^2}} \cdot (15x^2 + 2x)$

$h'(x) = \frac{15x^2 + 2x}{2\sqrt{5x^3+x^2}}$ ✓

Ex ⑥ Find the smallest positive value so that $f'(x) = 0$.

$$f(x) = \sin(x^2)$$

chain Rule

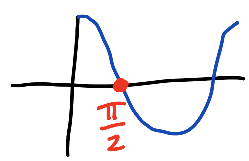
$$f'(x) = \cos(x^2) \cdot 2x$$

set it equal to 0

$$0 = \cos(x^2) \cdot 2x$$

$\cos(x^2) = 0$ or $2x = 0$

$x = 0$ is NOT positive



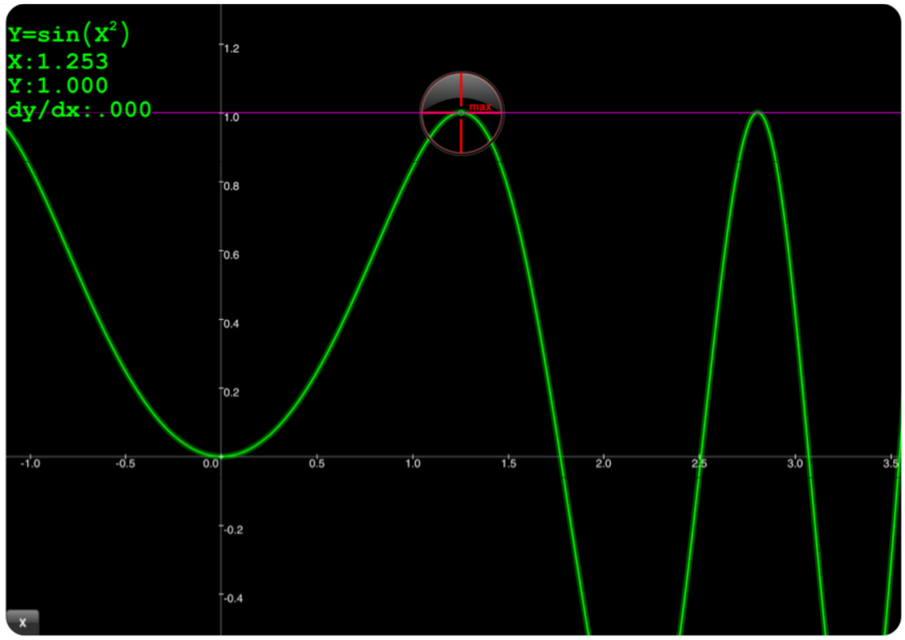
$$x^2 = \frac{\pi}{2}$$

$$x = \sqrt{\frac{\pi}{2}} \checkmark$$

Graph the original function to see if $x = \sqrt{\frac{\pi}{2}}$ looks right.

The function should have a horizontal tangent line when $x = \sqrt{\frac{\pi}{2}} = 1.253\ldots$

yes!



Ex ⑦ Which derivative rule or rules needs to be used?

A) $y = \sin(\tan x)$

B) $y = \sin x (\tan x)$

C) $y = \sin(\tan e^x)$

A) $y = \sin(\tan x)$ Chain Rule

$y' = \cos(\tan x) \cdot \sec^2 x$ ✓

B) $y = \sin x (\tan x)$ Product Rule

$y' = \cos x \cdot \tan x + \sin x \cdot \sec^2 x$ ✓

c) $y = \sin(\tan e^x)$ Chain Rule twice

$y' = \cos(\tan e^x) \cdot \sec^2(e^x) \cdot e^x$